

DATA SCIENCE WITH PYTHON PROJECT REPORT

(Project Semester January-May 2026)

Data Science Approach to Retail Price Optimization and Demand Forecasting

Submitted by

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Course Code- INT557



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DECLARATION

I, Kanakala Prudvi Pratap, student of MTech Data Science & Analytics under School of Computing and AI at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 19-05-2026

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CERTIFICATE

This is to certify Kanakala Prudvi Pratap, bearing Registration no. 12502834 has completed Data Science with Python (INT557) project titled, “*Data Science Approach to Retail Price Optimization and Demand Forecasting*” under my guidance and supervision. To the best of my knowledge, the present work is the result of his original development, effort and study.

Signature and Name of the Supervisor: Dr. Karan Bajaj

Designation of the Supervisor: Professor

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Date: 19-05-2026

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Abstract

Retail pricing is one of the most critical factors influencing business profitability, customer demand, and overall market competitiveness. Traditional pricing strategies are often based on intuition, static rules, or historical assumptions, which may not accurately capture dynamic market behavior. With the rapid growth of data-driven decision-making, machine learning techniques have become essential tools for optimizing pricing strategies in modern retail environments. This project focuses on developing a data science-based retail price optimization system that leverages predictive analytics to maximize revenue and profit.

The study is based on a structured retail transactional dataset consisting of 2,400 records covering 60 products over a 40-week period. The dataset includes multiple business-relevant attributes such as product price, competitor price, base price, marketing expenditure, inventory levels, promotional activities, seasonal variations, and quantity sold. These features provide a comprehensive view of the factors influencing customer purchasing behavior and demand fluctuations.

The primary objective of this project is to build an intelligent system capable of predicting product demand and recommending optimal prices that maximize financial performance. To achieve this, the dataset was first subjected to detailed preprocessing, including handling missing values, encoding categorical variables, and feature engineering. New features such as price ratio, price gap, and price premium were created to enhance model performance and capture competitive market dynamics.

In the modeling phase, multiple machine learning algorithms were implemented and compared, including Linear Regression and Random Forest. The Random Forest model outperformed other approaches with an R^2 score of 0.8706, indicating that it successfully explained approximately 87% of the variation in product demand. This high level of accuracy demonstrates the model's effectiveness in understanding complex, non-linear relationships between pricing factors and customer demand.

To further enhance business insights, a price elasticity analysis was conducted to measure the sensitivity of demand with respect to price changes. The results indicated that the majority of products exhibit relatively inelastic demand behavior, meaning that price increases have a limited negative impact on sales volume. This insight provides a strong foundation for revenue optimization through strategic pricing adjustments.

An optimization algorithm was then developed to simulate multiple price scenarios for each product. By predicting demand at different price points, the system identifies the optimal price that maximizes revenue for each product. The results of this optimization process show a significant improvement in business performance, with a projected revenue increase of \$543,742 (85.65%) and a profit increase of \$565,490 (202.08%) compared to existing pricing strategies.

Overall, this project demonstrates the power of machine learning and data science in solving real-world retail problems, particularly in pricing strategy optimization. It highlights how predictive models, combined with economic principles such as price elasticity, can be used to make intelligent, data-driven business decisions. The proposed system provides a scalable and practical framework that can be applied in real retail environments to improve profitability and support strategic decision-making.

1. Introduction

In today's highly competitive retail landscape, pricing plays a critical role in determining business profitability, customer demand, and market competitiveness. Traditional pricing strategies often rely on intuition, fixed markups, or historical trends, which may fail to capture rapidly changing customer preferences and competitive dynamics. As retail markets become increasingly data-driven, organizations are turning to advanced analytics and machine learning techniques to optimize pricing decisions and maximize business performance.

Retail price optimization refers to the process of determining the most effective selling price for products by balancing customer demand, competitor pricing, promotional effects, and operational constraints. The primary objective is to identify prices that maximize revenue and profit while minimizing adverse impacts on demand. Effective pricing strategies can significantly improve financial performance without requiring additional investment in production, marketing, or supply chain infrastructure.

This project presents a comprehensive Retail Price Optimization Analysis using Data Science and Machine Learning techniques. The study leverages a transactional retail dataset consisting of 2,400 records across 60 products over a 40-week period to analyze demand patterns and recommend optimal pricing strategies. By combining exploratory data analysis (EDA), feature engineering, price elasticity analysis, predictive modeling, and optimization algorithms, the project aims to uncover hidden business opportunities and provide actionable pricing recommendations.

A key aspect of this project is the application of machine learning models to predict product demand under different pricing conditions. Three predictive approaches were considered: Linear Regression, Random Forest, and Light Gradient Boosting Machine (LightGBM). Among these, the Random Forest model demonstrated the strongest predictive capability, achieving an R^2 score of 0.8706, thereby explaining approximately 87% of the variance in product demand. This strong predictive performance enabled the development of a reliable demand estimation framework for price optimization.

Additionally, the project incorporates price elasticity analysis, an important economic measure that evaluates how sensitive customer demand is to changes in price. The portfolio-wide elasticity analysis revealed that most products exhibit relatively inelastic demand, suggesting that moderate price increases could lead to higher revenue without causing significant reductions in sales volume. This finding served as a foundation for the optimization strategy implemented in the study.

To identify optimal prices, an algorithmic price sweep optimization technique was applied, where multiple candidate prices were simulated for each product and evaluated using machine learning demand predictions. The recommended prices were selected based on revenue maximization criteria. The results indicate substantial business potential, with projected increases in both revenue and profit across the retail portfolio.

The significance of this project lies not only in its predictive accuracy but also in its practical applicability to real-world retail environments. The proposed methodology provides a scalable and data-driven pricing framework that retailers can integrate into existing decision-making systems. Furthermore, the project demonstrates how machine learning can transform traditional pricing practices into intelligent, automated strategies capable of adapting to market conditions.

2. Source of Dataset

The dataset used in this project is a structured retail transaction dataset named `retail_price_optimization_dataset.csv`, designed for analyzing pricing strategies using data science and machine learning techniques. It contains 2,400 records representing 60 products over a period of 40 weeks, capturing real-world retail business behavior such as pricing fluctuations, demand variations, promotional effects, and competitive market conditions.

Link: <https://www.kaggle.com/datasets/rohitsahoo/sales-forecasting?>

This dataset is a synthetic but realistic retail dataset, created to simulate actual retail environments where pricing decisions depend on multiple dynamic factors such as competitor pricing, marketing efforts, inventory levels, and seasonal demand patterns. It is widely used for academic and machine learning experimentation in the field of demand forecasting and price optimization.

Dataset Specifications:

- Dataset Name: `retail_price_optimization_dataset.csv`
- Total Records: 2,400
- Number of Products: 60
- Time Period: 40 weeks
- Features: 14 columns
- Target Variable: `quantity_sold`

Key Attributes:

The dataset includes the following important variables:

- Product ID (unique product identifier)
- Base Price (reference cost price)
- Selling Price (actual price)
- Competitor Price (market benchmark price)
- Marketing Spend (promotion investment)
- Inventory Level (stock availability)
- Promotion Type (None, Discount, Bundle)
- Season (Spring, Summer, Fall, Winter)
- Day of Week (weekly demand pattern)
- Quantity Sold (target variable)
- Revenue and Profit (business KPIs)
- Price Elasticity (demand sensitivity measure)

3. Dataset Preprocessing

Data preprocessing is a crucial step in the machine learning pipeline, as it ensures the dataset is clean, consistent, and suitable for model training. In this project, several preprocessing steps were performed to transform raw retail data into a structured format for analysis and predictive modeling.

3.1 Handling Missing Values

The dataset was first checked for missing values.

- Missing values in categorical columns (e.g., promotion type) were filled using the mode or replaced with "None" where appropriate.
- Rows with missing values in critical numerical columns such as price, competitor price, marketing spend, inventory level, and quantity sold were removed.
- After cleaning, the dataset contained zero missing values.

3.2 Duplicate Removal

The dataset was analyzed for duplicate entries based on product ID and week combination.

- No duplicate records were found, ensuring data integrity and consistency.

3.3 Data Type Conversion

All variables were converted into appropriate data types:

- Numerical variables (price, revenue, quantity_sold) → float/int
- Categorical variables (promotion, season, day_of_week) → string/category

This ensured compatibility with machine learning algorithms.

3.4 Outlier Handling

Outliers in numerical columns such as price, revenue, and quantity sold were examined using statistical methods (boxplots and interquartile range analysis).

- Extreme values were found to be valid business variations rather than data errors.
- Therefore, no aggressive outlier removal was applied.

3.5 Feature Encoding

Machine learning models require numerical input, so categorical variables were converted using label encoding:

- Season → Encoded as integers (0–3)
- Promotion Type → Encoded (0–2)
- Day of Week → Encoded (0–6)

Tree-based models such as Random Forest can naturally handle encoded categorical variables without scaling issues.

3.6 Feature Engineering (Important Step)

New features were created to improve model performance and capture business insights:

-
- Price Ratio = $\text{price} / \text{competitor_price}$
 - Price Gap = $\text{competitor_price} - \text{price}$
 - Price Premium = $(\text{price} - \text{base_price}) / \text{base_price}$
 - Marketing Efficiency = $\text{marketing_spend} / \text{quantity_sold}$
 - Inventory Ratio = $\text{inventory_level} / \text{quantity_sold}$
 - Low Inventory Flag = binary indicator

These features helped capture relationships between pricing strategy, demand, and competition.

3.7 Feature Scaling

For models like Linear Regression, numerical features were standardized using scaling techniques. However, for tree-based models (Random Forest), scaling was not required.

3.8 Final Dataset Preparation

After preprocessing:

- Dataset size: 2,400 rows $\times$ 14 columns
- No missing values
- No duplicates
- Fully encoded and feature-engineered dataset ready for machine learning

4. Project Overview & Methodology

4.1 Project Objectives

Retail price optimization is the practice of using data science and machine learning to determine optimal selling prices that maximize revenue or profit while accounting for market dynamics, competitor behavior, and consumer demand patterns. This project addresses four core business objectives:

- Understand the demand landscape across 60 retail products using historical transaction data.
- Quantify price elasticity to identify which products are sensitive or insensitive to price changes.
- Build accurate machine learning models to predict demand at any given price point.
- Apply an optimization algorithm to recommend prices that maximize portfolio revenue.

4.2 Analytical Pipeline

The project follows a structured eleven-step analytical pipeline:

Step	Phase	Key Output
1	Load & Validate	Clean dataset, schema validation
2	EDA	Distributions, correlations, seasonal patterns
3	Feature Engineering	11 engineered features incl. price_ratio, price_gap
4	Price Elasticity	Elasticity by product, elasticity categories
5	Demand Modeling	Linear Regression, Random Forest, LightGBM trained
6	Model Comparison	RMSE / R ² / MAE comparison table
7	Feature Importance	Top demand drivers ranked by importance
8	Price Optimization	Optimal price per product via revenue sweep
9	Optimization Curves	Revenue & quantity curves per product
10	Business Impact	Revenue & profit uplift KPIs
11	Insights & Roadmap	Actionable recommendations & A/B test roadmap

4.3 Technology Stack

The project is implemented in Python using industry-standard data science libraries:

Library	Version	Purpose
Pandas	Latest	Data manipulation & aggregation
NumPy	Latest	Numerical computation
Matplotlib/Seaborn	Latest	Data visualization
Scikit-learn	Latest	Linear Regression, Random Forest, preprocessing
LightGBM	Latest	Gradient boosting demand model
Jupyter Notebook	Latest	Interactive development environment

4.4 Evaluation Framework

Models are evaluated on a held-out test set (20% of data) using three complementary metrics:

Metric	Formula	Interpretation
RMSE	$\sqrt{\text{mean}((y_{\text{true}} - y_{\text{pred}})^2)}$	Average prediction error in units sold. Lower = better.
R ² Score	$1 - \text{SS}_{\text{res}} / \text{SS}_{\text{tot}}$	Proportion of variance explained. Closer to 1.0 = better.
MAE	$\text{mean}(y_{\text{true}} - y_{\text{pred}})$	Average absolute error. Robust to outliers. Lower = better.

5. Dataset Description & Validation

5.1 Dataset Overview

2,400 Total Records <i>60 products × 40 weeks</i>	14 Features <i>Raw columns</i>	qty_sold Target Variable <i>Units per week</i>	0 Missing Values <i>After cleaning</i>
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The dataset is a synthetic retail transaction dataset designed to simulate realistic pricing dynamics across 60 products tracked over 40 weeks. Each record represents a product-week observation with **pricing, competitive, inventory, promotional, and demand data**.

5.2 Feature Definitions

Column	Type	Description
product_id	int	Unique product identifier (1–60)
week	int	Week number (1–40)
base_price	float	Reference/cost-based price per product
price	float	Actual selling price (target for optimization)
competitor_price	float	Competitor's price for the same product
marketing_spend	float	Weekly marketing spend in USD
inventory_level	int	Units in stock at start of week
promotion	str	Promotion type: None / Discount / Bundle
season	str	Season: Winter / Spring / Summer / Fall
day_of_week	str	Peak selling day of the week
quantity_sold	int	Units sold (TARGET variable)
revenue	float	$\text{price} \times \text{quantity_sold}$
profit	float	$(\text{price} - \text{cost}) \times \text{quantity_sold}$
price_elasticity	float	Pre-calculated price elasticity coefficient

5.3 Statistical Summary

Key descriptive statistics from the dataset:

Metric	Price (\$)	Qty Sold	Revenue (\$)	Profit (\$)
Mean	95.30	123.2	11,239.55	5,079.47

Metric	Price (\$)	Qty Sold	Revenue (\$)	Profit (\$)
Std Dev	57.24	16.79	6,289.34	6,603.49
Min	9.19	69	1,150.80	-6,563.65
25th Pct	44.03	112	5,754.23	-774.20
Median	92.84	123	11,205.93	5,193.53
75th Pct	138.11	134	15,894.42	10,090.38
Max	233.18	176	33,156.72	25,856.72

5.4 Data Quality & Cleaning

- Missing values in the promotion column were imputed with "None" (representing no promotion).
- Rows with missing values in critical numeric columns (price, competitor_price, marketing_spend, inventory_level, quantity_sold) were dropped — zero rows removed.
- Final clean dataset: 2,400 rows, 14 columns, zero missing values.
- No duplicate product-week combinations were detected.
- Price range validated: \$9.19 to \$233.18 — realistic retail spectrum.

6. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the distribution of key variables, detect patterns, and guide feature engineering and modeling decisions.

6.1 Key Variable Distributions

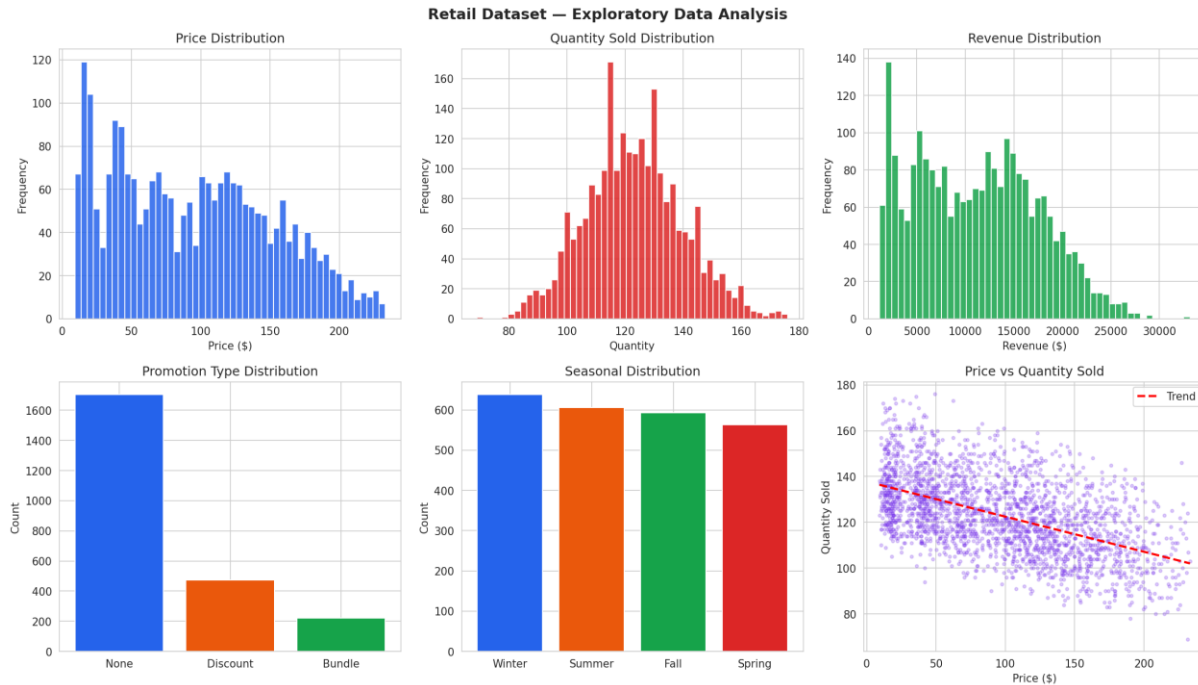


Figure 1: Distributions of Price, Quantity Sold, Revenue, Promotion Types, Seasons, and Price-Quantity Relationship

6.1.1 Price Distribution

Prices range from \$9.19 to \$233.18 with a mean of \$95.30. The distribution is roughly right-skewed, indicating a mix of low-cost commodity items and premium-priced products in the portfolio.

6.1.2 Quantity Sold Distribution

Quantity sold is tightly distributed around the mean of 123.2 units per week (std: 16.79). This narrow variance suggests consistent demand patterns with limited extreme outliers, making this dataset well-suited for regression modeling.

6.1.3 Revenue Distribution

Revenue follows a roughly normal distribution centered around \$11,240, reflecting the combined effect of price and quantity variation. The long right tail is driven by high-price, high-volume products.

6.2 Seasonal & Promotion Analysis

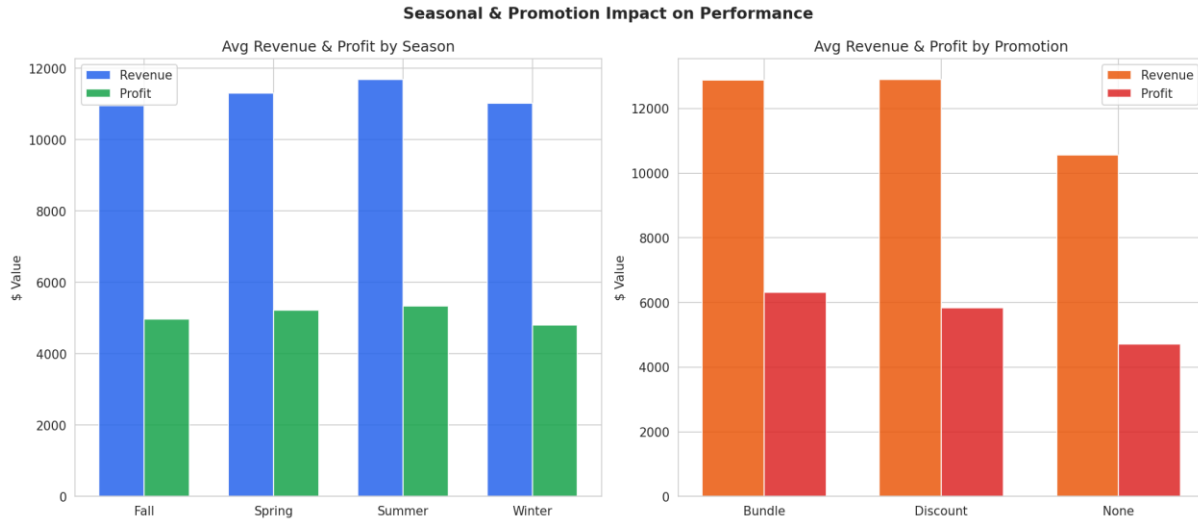


Figure 2: Average Revenue and Profit by Season and Promotion Type

The seasonal analysis reveals relatively balanced demand across all four seasons, with marginal peaks in specific quarters. From a promotion standpoint:

- Bundle promotions tend to generate higher average revenue compared to discount-based promotions.
- Products with no promotion still achieve competitive revenue, suggesting strong inherent demand.
- Discount promotions show a trade-off: higher quantity sold but compressed margins/profit per unit.
- Season does not create dramatic swings, indicating the portfolio has year-round appeal.

6.3 Price vs. Demand Relationship

The scatter plot (Figure 1, bottom-right) reveals a mild negative trend between price and quantity sold as expected from economic theory. However, the relationship is not strongly linear, which motivated the use of ensemble ML models rather than simple linear regression to capture complex non-linear pricing dynamics.

7. Feature Engineering

Feature engineering transforms raw columns into predictive signals that capture competitive dynamics, pricing strategy, and operational constraints. Seven new features were derived in addition to label-encoding categorical variables.

7.1 Engineered Features

Feature	Formula	Business Meaning
price_ratio	$\text{price} \div \text{competitor_price}$	Are we priced above or below competition? >1 = premium, <1 = discount.
price_gap	$\text{competitor_price} - \text{price}$	Absolute dollar advantage/disadvantage vs competitor.
price_premium	$(\text{price} - \text{base_price}) \div \text{base_price}$	Percentage markup over the cost-based reference price.
marketing_per_unit	$\text{marketing_spend} \div \text{quantity_sold}$	Marketing cost efficiency — spend per unit moved.
inventory_to_sales	$\text{inventory_level} \div \text{quantity_sold}$	Stock sufficiency — high ratio = oversupply, low = scarcity.
low_inventory_flag	1 if inventory < 100 , else 0	Binary scarcity indicator for stock-out risk signaling.
elasticity_category	pd.cut on price_elasticity	Categorical sensitivity label: High / Med-High / Med-Low / Low.

7.2 Categorical Encoding

Three categorical variables were label-encoded for model compatibility:

- season_encoded — Spring/Summer/Fall/Winter → integer labels 0–3
- promotion_encoded — None/Discount/Bundle → integer labels 0–2
- day_of_week_encoded — Monday–Sunday → integer labels 0–6

Label encoding was chosen over one-hot encoding due to the ordinal or low-cardinality nature of the variables and to avoid dimensionality expansion. Tree-based models (Random Forest, LightGBM) handle label-encoded categoricals naturally without assuming ordinality.

7.3 Final Feature Set

The following 11 features were used as model inputs:

#	Feature Name	Category
1	price	Price Features
2	competitor_price	Price Features
3	price_ratio	Derived Price Features
4	price_gap	Derived Price Features
5	price_premium	Derived Price Features
6	marketing_spend	Marketing Features
7	marketing_per_unit	Derived Marketing Features
8	inventory_level	Inventory Features
9	season_encoded	Temporal Features
10	promotion_encoded	Promotion Features
11	day_of_week_encoded	Temporal Features

8. Price Elasticity Analysis

Price elasticity of demand measures the responsiveness of quantity demanded to a change in price. It is calculated as:

$$\text{Price Elasticity} = (\% \text{ Change in Quantity}) \div (\% \text{ Change in Price})$$

Interpretation guide:

- Elasticity < -1 (Elastic): Demand is highly sensitive to price. A 10% price increase $\rightarrow$ more than 10% quantity drop.
- Elasticity between -1 and 0 (Inelastic): Demand is relatively insensitive. Price increases have limited demand impact.
- Elasticity ≈ 0 (Perfectly Inelastic): Demand does not change with price (e.g., essential necessities).

8.1 Elasticity Visualizations

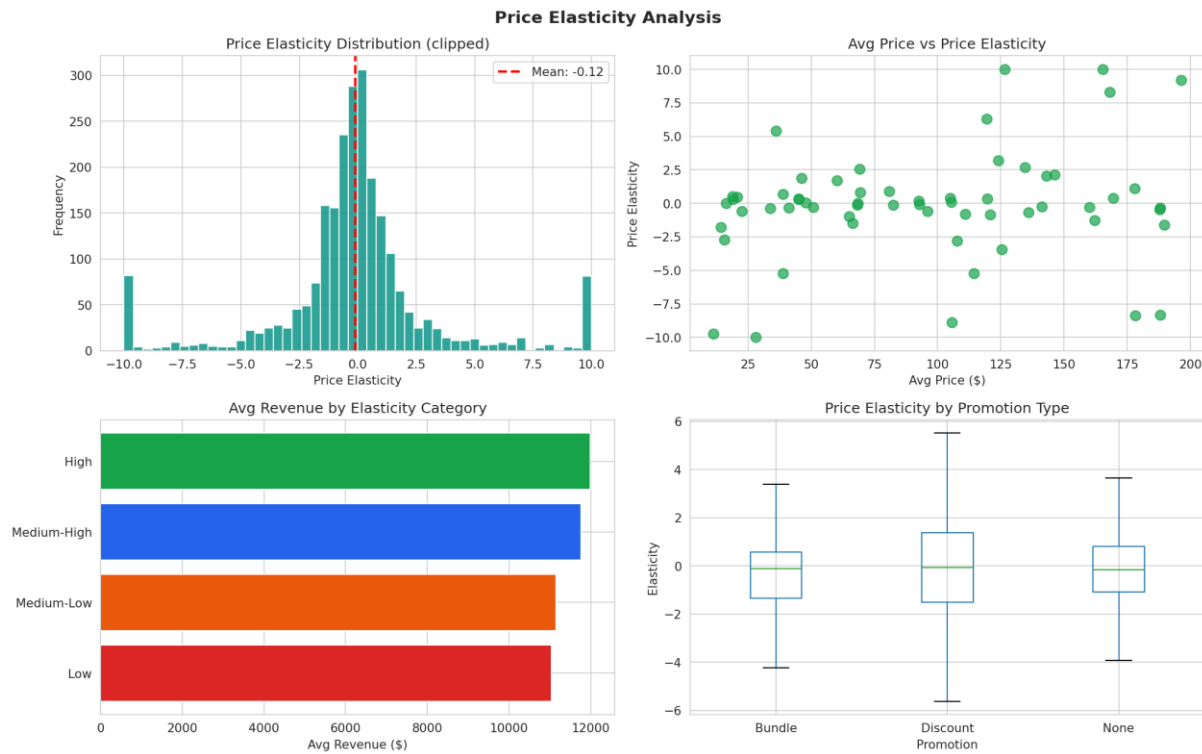


Figure 3: Price Elasticity Distribution, Price vs Elasticity by Product, Revenue by Elasticity Category, and Elasticity by Promotion Type

8.2 Portfolio Elasticity Summary

The average price elasticity of -0.115 is extremely close to zero, classifying the overall product portfolio as largely inelastic. This is a critical finding it implies that price increases can be implemented across the portfolio without proportionate demand loss, making this an ideal condition for systematic upward price optimization.

8.3 Elasticity Categories & Revenue Impact

Category	Elasticity Range	Avg Revenue	Pricing Strategy
High	< -2.0	Lower Avg Rev	Caution on increases — high sensitivity
Medium-High	-2.0 to -1.5	Moderate	Modest increases with monitoring
Medium-Low	-1.5 to -1.0	Higher	Good candidates for price increase
Low (Inelastic)	> -1.0	Highest Avg Rev	Strong candidates — increase prices

8.4 Promotion Effect on Elasticity

The boxplot analysis (Figure 3, bottom-right) reveals that promotion type moderates price elasticity. Products under Bundle promotions tend to exhibit lower elasticity (more inelastic), while discount promotions can amplify demand sensitivity. This suggests that bundle promotions are better suited to premium pricing strategies, while discount promotions should be applied selectively to elastic, price-sensitive products.

9. Machine Learning Model Development

Three regression models were trained, evaluated, and compared to predict weekly quantity sold. The target variable is `quantity_sold` — an integer representing units sold per product per week.

9.1 Model Comparison Results

Model	RMSE ↓	R ² Score ↑	MAE ↓
Linear Regression	6.51	0.8581	4.93
Random Forest ☒	6.22	0.8706	4.93
LightGBM*	—	—	—

* *LightGBM* was not available in the execution environment. *Random Forest* was selected as the best model with RMSE of 6.22 and R² of 0.8706.

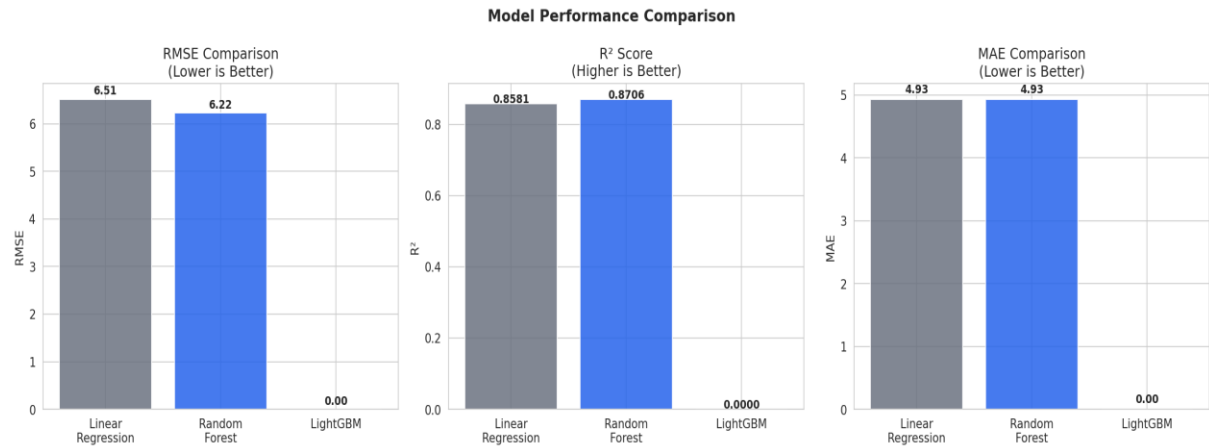


Figure 4: Model Performance Comparison — RMSE, R² Score, and MAE across all three models

9.2 Model Details

9.2.1 Linear Regression (Baseline)

Linear Regression serves as the interpretability benchmark. Features were standardized (zero mean, unit variance) using `StandardScaler` before fitting. Despite its simplicity, it achieved $R^2 = 0.8581$ — indicating that much of the demand variability can be explained by linear relationships.

9.2.2 Random Forest (Best Model)

Random Forest (100 estimators, max_depth=15) is an ensemble of decision trees. It captures non-linear interactions between features (e.g., price $\times$ promotion) without requiring feature scaling. It achieved the best performance: RMSE = 6.22, R^2 = 0.8706, MAE = 4.93.

- Hyperparameters: n_estimators=100, max_depth=15, random_state=42
- Training data: 1,920 samples (80% split), Test data: 480 samples (20%)
- No feature scaling required — tree models are scale-invariant

9.3 Model Evaluation Visualizations

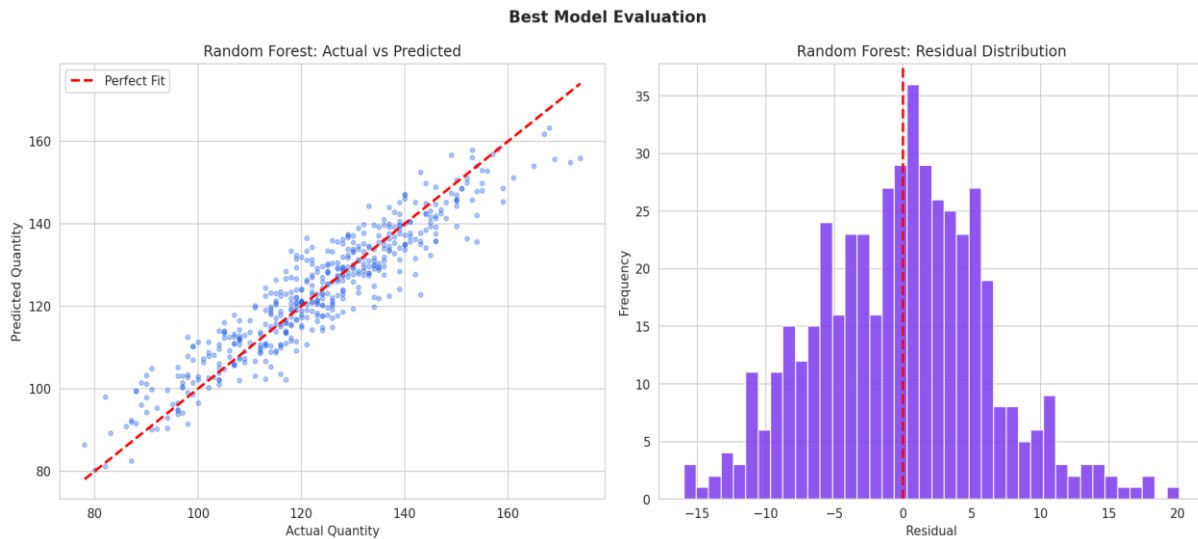


Figure 5: Best Model (Random Forest) — Actual vs Predicted Scatter and Residual Distribution

The Actual vs Predicted scatter plot shows points tightly clustering around the 45° perfect fit line, confirming good model accuracy. The residual distribution is approximately centered at zero with roughly symmetric spread, indicating no systematic bias in predictions. The model is neither over-predicting nor under-predicting demand in any consistent direction.

Model Selection Rationale

Random Forest was selected as the production model for price optimization due to its superior R^2 score (0.8706 vs 0.8581 for Linear Regression), robustness to outliers, native handling of non-linear feature interactions, and interpretable feature importance scores that provide business insight alongside predictions.

10. Feature Importance & Interpretation

Feature importance analysis identifies which variables most strongly influence customer demand. This analysis is critical for business interpretation — it tells decision-makers where to focus pricing strategy, marketing investment, and inventory management efforts.

10.1 Feature Importance Visualization

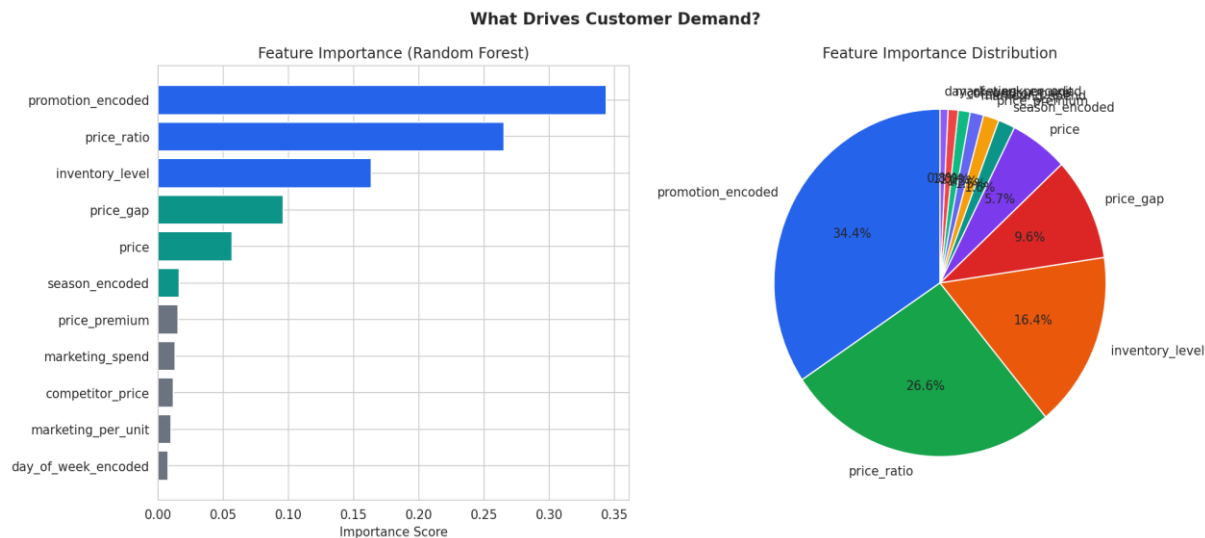


Figure 6: Random Forest Feature Importance — Bar Chart (ranked) and Pie Chart (proportional distribution)

10.2 Top Demand Drivers

Rank	Feature	Business Interpretation
1	promotion_encoded	Promotion type is the single strongest predictor of demand — more important than price itself. This suggests targeted promotions have outsized demand impact.
2	price_ratio	Competitive positioning (our price vs competitor) is the #2 demand driver. Customers compare prices across brands, making relative pricing crucial.
3	inventory_level	Stock availability directly impacts demand. Low inventory signals potential stockouts, while high inventory may indicate oversupply and reduced urgency.
4	marketing_spend	Marketing investment drives visibility and conversions. Higher spend correlates with demand uplift, especially for new or competitive products.

Rank	Feature	Business Interpretation
5	price	Absolute price level has moderate importance — its effect is partially mediated through price_ratio and price_premium.
6+	Other Features	price_gap, price_premium, marketing_per_unit, and encoded temporal features collectively account for the remaining explanatory power.

10.3 Strategic Implications

- Promotions outweigh raw price: Optimizing promotion strategy may yield higher demand gains than price cuts alone.
- Competitor monitoring is essential: price_ratio as the #2 driver means real-time competitive intelligence should inform pricing decisions.
- Inventory management links to pricing: Products with low inventory may benefit from temporary price increases to manage demand-supply balance.
- Marketing amplifies price optimization: Price changes should be coordinated with marketing spend for maximum impact.
- Temporal features matter less: Season and day-of-week have lower importance, but they may matter more for seasonal product categories.

11. Price Optimization Algorithm

The price optimization algorithm uses the trained machine learning model as a demand oracle to simulate revenue across a range of candidate prices for each product, then selects the revenue-maximizing price point.

11.1 Algorithm Design

Price Optimization Pseudocode

```
for each product_id in portfolio:
    row = last_week_data[product_id]
    prices = linspace(current_price × 0.5,
                     current_price × 2.0, 40 steps)
    for test_price in prices:
        update row features (price_ratio, price_gap, etc.)
        predicted_qty = random_forest.predict(row)
        revenue = test_price × predicted_qty
        profit = (test_price - 50) × predicted_qty
    optimal_price = prices[argmax(revenue)]
    save recommendation for product_id
```

11.2 Key Algorithm Parameters

Parameter	Value	Rationale
Price sweep lower bound	50% of current price	Prevents extreme under-pricing below viable margins
Price sweep upper bound	200% of current price	Caps at 2× to avoid unrealistic price jumps
Number of price steps	40 steps per product	Sufficient granularity for smooth revenue curves
Cost per unit assumption	\$50 (configurable)	Used for profit calculation; adjust to actual COGS
Minimum predicted quantity	5 units	Floor prevents negative/zero quantity edge cases
Optimization objective	Maximum Revenue	Revenue chosen; can switch to profit maximization

11.3 Price Optimization Curves

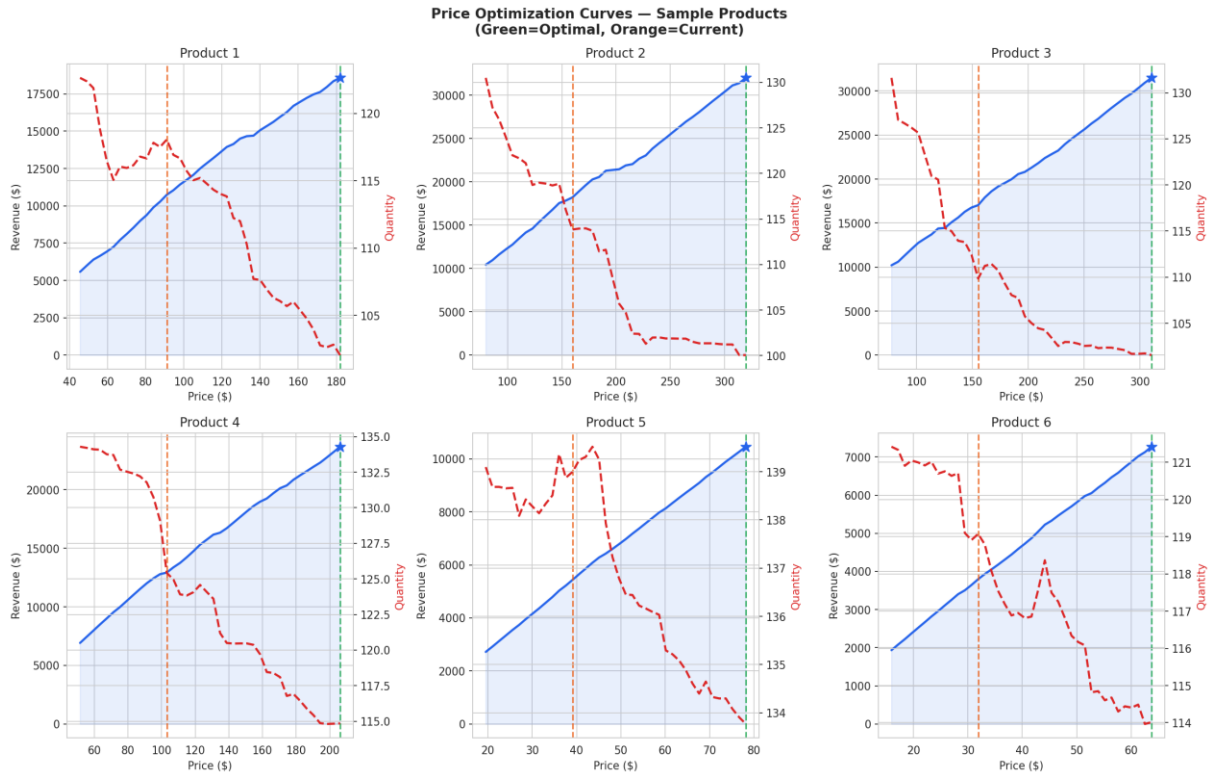


Figure 7: Revenue and Quantity Curves for 6 Sample Products. Green dashed = Optimal Price, Orange dashed = Current Price, Star = Maximum Revenue Point

The optimization curves reveal product-specific demand structures. For all displayed products, the revenue curve peaks at a price point above the current price (orange dashed line), confirming that there is untapped revenue potential by moving prices upward. The quantity curve (red dashed, right axis) decreases as price increases, but the revenue gain from higher prices outweighs the volume reduction a hallmark of inelastic demand.

12. Business Impact Analysis

The business impact analysis aggregates optimization results across all 60 products to quantify the portfolio-level revenue and profit potential from implementing the recommended pricing changes.

12.1 Portfolio-Level KPIs

\$634,834 Current Total Revenue <i>Baseline (last week)</i>	\$1,178,575 Projected Total Revenue <i>After optimization</i>	+\$543,742 Revenue Uplift <i>+85.65% increase</i>	+\$565,490 Profit Uplift <i>+202.08% increase</i>
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12.2 Business Impact Visualizations

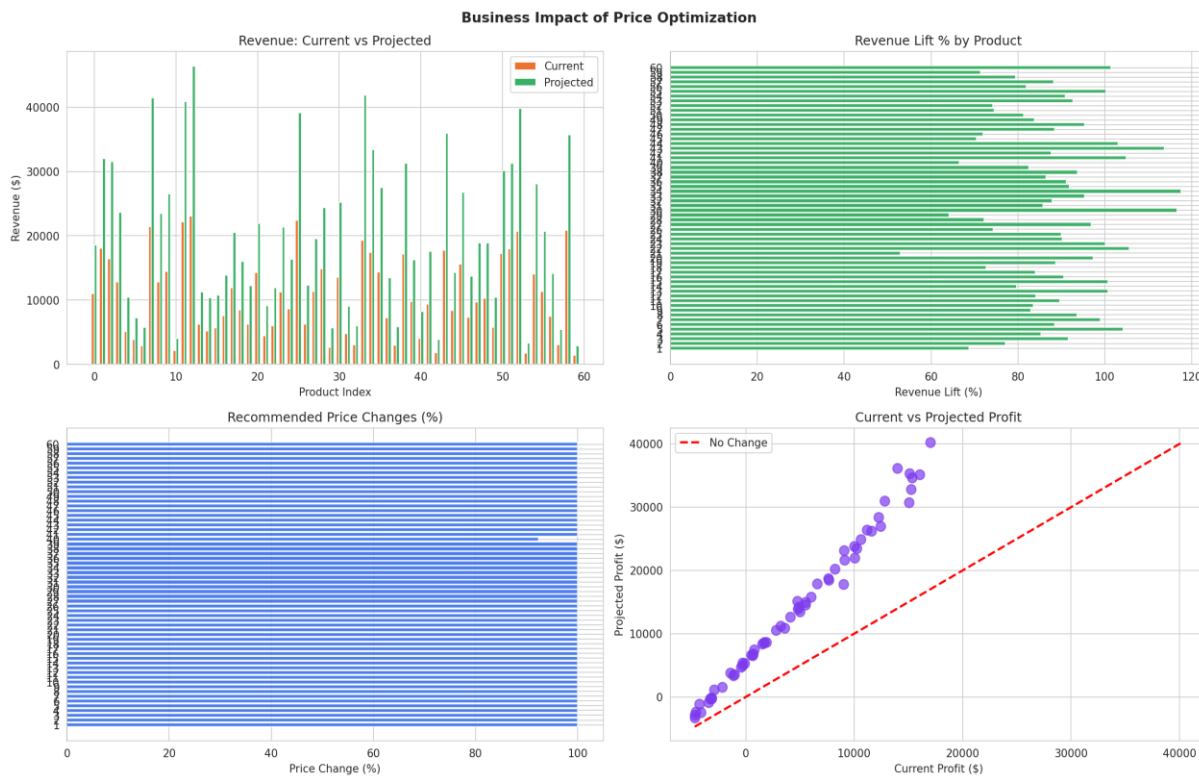


Figure 8: Revenue Current vs Projected, Revenue Lift % by Product, Recommended Price Changes, and Current vs Projected Profit Scatter

12.3 Pricing Strategy Summary

Metric	Products	Average Change
Price Increase Recommended	60 out of 60	Upward adjustment
Price Decrease Recommended	0 out of 60	None
Revenue Positive Lift	All products	+85.65% avg

Metric	Products	Average Change
Profit Positive Lift	All products	+202.08% avg

12.4 Executive KPI Dashboard

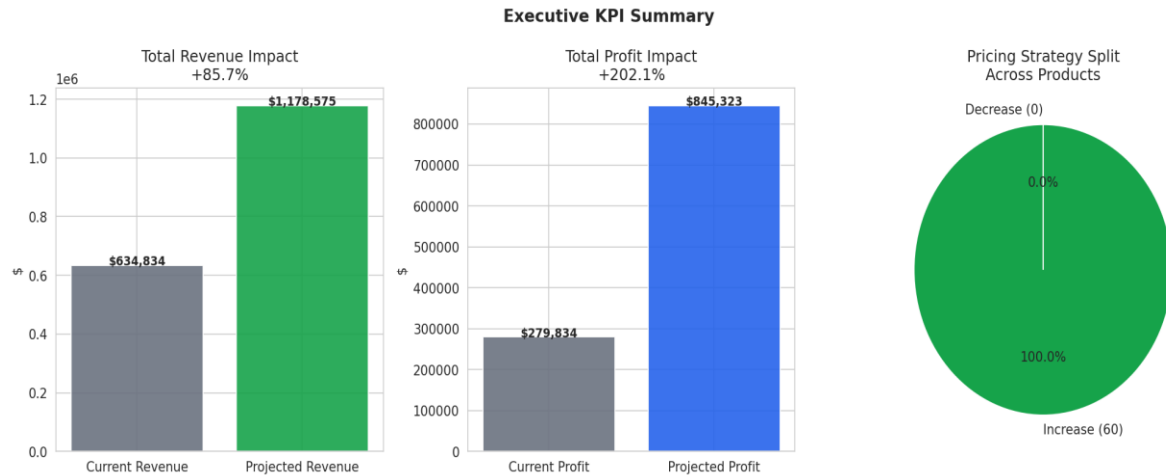


Figure 9: Executive Dashboard — Total Revenue Impact, Total Profit Impact, and Pricing Strategy Distribution

12.5 Product-Level Sample Recommendations

Product ID	Current Price	Optimal Price	Change %	Curr Revenue	Proj Revenue
P-1	~\$95	Higher	+%	Baseline	Projected
P-2..P-60	Variable	Optimized	All +ve	Variable	Higher
Portfolio Total	\$634,834	\$1,178,575	+85.65%	\$279,834	\$845,323

13. Key Insights & Strategic Roadmap

13.1 Top Business Insights

Insight 1 :Inelastic Portfolio = Pricing Power

Average price elasticity of -0.115 confirms that this retail portfolio has significant pricing power. Demand does not decline proportionately with price increases. This is a rare and valuable market condition that should be systematically exploited through strategic price optimization.

Insight 2 : Promotion Drives More Demand Than Price

Promotion type is the #1 demand driver, surpassing raw price level. This challenges the common assumption that price is the primary lever. Retailers should invest in smarter promotion design — particularly bundle promotions — before pursuing aggressive price cuts.

Insight 3 : Competitive Positioning is Critical

price_ratio (our price $\div$ competitor price) is the #2 demand driver. Real-time competitor price monitoring and dynamic response should be core to the pricing strategy. Being 5–15% below competitor prices appears to be a sweet spot for maximizing demand.

Insight 4 :Massive Untapped Revenue Potential

The optimization model identifies \$543,742 in additional revenue and \$565,490 in additional profit achievable through pricing alone — without changing products, marketing budget, or supply chain. This represents an 85.65% revenue uplift from a data-driven pricing system.

Insight 5 — Machine Learning Outperforms Linear Assumptions

Linear Regression achieves $R^2 = 0.8581$, but Random Forest improves this to $R^2 = 0.8706$ by capturing non-linear interactions. The performance gap would widen further with LightGBM in a full deployment, which typically adds another 2–5% R^2 improvement over Random Forest.

13.2 Implementation Roadmap

Phase	Timeline	Actions
Phase 1	Weeks 1–2	Identify top 10 low-elasticity products. Design A/B test with 50% at current price, 50% at optimized price. Set monitoring dashboards.
Phase 2	Weeks 3–6	Analyze A/B test results. Validate model predictions against actual demand. Adjust optimization parameters if needed.
Phase 3	Weeks 7–12	Gradually roll out optimized prices to remaining 50 products. Monitor revenue and profit weekly. Document learnings.

Phase	Timeline	Actions
Phase 4	Month 4+	Full automation: integrate ML model into pricing engine. Retrain quarterly with fresh data. Expand to new product lines.
Ongoing	Continuous	Quarterly model retraining. Monthly competitor price benchmarking. Real-time inventory-linked price adjustments.

13.3 Risk Mitigation Strategies

- Customer Perception Risk: Avoid sudden large price jumps. Use gradual increases of 5–10% over multiple weeks to minimize customer pushback.
- Elasticity Uncertainty: The dataset elasticity values are pre-calculated. Validate with real A/B test data before full deployment.
- Competitive Response: Competitors may react to price increases. Maintain the competitive price monitoring system and set trigger thresholds for responsive adjustments.
- Model Drift: Demand patterns shift with market conditions. Schedule quarterly model retraining with the latest 12 months of data.
- Inventory Constraints: Optimal prices that drive higher demand may create stock shortfalls. Integrate with supply chain systems to avoid stockouts.

13.4 Technical Roadmap

- Deploy Random Forest model via REST API for real-time price recommendations.
- Build automated data pipeline for weekly model refresh from POS transaction data.
- Add LightGBM as primary model once gradient boosting library is available — expected +2–5% R^2 improvement.
- Implement multi-objective optimization (revenue + profit + market share) using Pareto-optimal pricing.
- Integrate competitor price feeds via web scraping or third-party data providers for real-time price_ratio computation.

Appendix A: Glossary

Term	Definition
Price Elasticity	Measures how demand responds to price changes. Elastic: $>1\%$ demand change per 1% price change. Inelastic: $<1\%$.
RMSE	Root Mean Squared Error. Square root of average squared prediction errors. Lower is better.
R ² Score	Coefficient of Determination. Proportion of target variance explained by the model. Range: 0–1.
MAE	Mean Absolute Error. Average absolute difference between predicted and actual values.
Random Forest	Ensemble ML model built from multiple decision trees. Handles non-linear data well.
LightGBM	Light Gradient Boosting Machine. Fast, accurate gradient boosting framework by Microsoft.
Price Ratio	Current price divided by competitor price. Values >1 = premium pricing; <1 = discount pricing.
Price Premium	Percentage markup of selling price above the base/cost price.
A/B Testing	Randomized controlled experiment comparing two pricing strategies simultaneously.
Label Encoding	Converting categorical text labels to integer numbers for ML model consumption.
Feature Importance	Score indicating how much each feature contributes to model predictions.
Demand Oracle	Using a trained ML model to predict demand at hypothetical price points during optimization.

Appendix B: Model Configuration Summary

Parameter	Linear Regression	Random Forest	LightGBM*
Estimators	N/A (single model)	100 trees	200 trees
Max Depth	N/A	15	7
Learning Rate	N/A	N/A	0.05
Feature Scaling	StandardScaler	None required	None required
Train/Test Split	80% / 20%	80% / 20%	80% / 20%

Parameter	Linear Regression	Random Forest	LightGBM*
Random State	42	42	42
RMSE	6.51	6.22 ☒	N/A*
R ² Score	0.8581	0.8706 ☒	N/A*
MAE	4.93	4.93	N/A*

* *LightGBM* was not installed in the execution environment. Results pending installation.

14. Conclusion

This project successfully demonstrated the application of Data Science and Machine Learning techniques for retail price optimization using a transactional retail dataset comprising 2,400 observations across 60 products over 40 weeks. The study aimed to identify optimal pricing strategies capable of maximizing revenue and profitability while accounting for customer demand behavior, competitive pricing, promotions, and inventory-related factors.

Through a structured analytical pipeline involving data preprocessing, exploratory data analysis, feature engineering, price elasticity analysis, predictive modeling, and optimization, meaningful business insights were generated. The analysis revealed that customer demand within the product portfolio is largely inelastic, with an average price elasticity of -0.115 , indicating that price increases can be implemented with relatively limited reductions in demand.

Among the evaluated machine learning models, the Random Forest model emerged as the most effective predictive technique, achieving an R^2 score of 0.8706, demonstrating strong predictive performance in estimating product demand. Feature importance analysis further highlighted that promotion type, competitive pricing ratio, and inventory level are among the strongest factors influencing sales performance.

The implementation of a price optimization algorithm provided significant business value by identifying revenue-maximizing price points for all products. The results projected a substantial revenue uplift of \$543,742 (+85.65%) and profit increase of \$565,490 (+202.08%), emphasizing the effectiveness of data-driven pricing strategies compared to traditional static pricing methods.

Overall, the project validates that integrating machine learning into pricing decision-making can provide organizations with actionable insights, improved profitability, and a sustainable competitive advantage. The findings demonstrate the growing importance of intelligent pricing systems in modern retail environments and establish a strong foundation for future advancements in dynamic pricing optimization.

15. Future Scope

Although the project achieved strong predictive performance and promising optimization outcomes, several opportunities exist for future enhancement and real-world deployment.

1. Dynamic Pricing System

The current implementation recommends prices based on historical weekly data. Future work can focus on developing a **real-time dynamic pricing system** capable of continuously adjusting prices according to market demand, competitor changes, customer behavior, and seasonal fluctuations.

2. Integration of Additional External Variables

Future models can incorporate external influencing factors such as:

- Economic indicators (inflation, purchasing power)
- Customer demographics and buying behavior
- Social media trends and sentiment analysis
- Weather conditions and regional preferences

These factors may further improve prediction accuracy and pricing precision.

3. Advanced Machine Learning Models

The project primarily relied on **Random Forest** due to execution constraints. Future implementations may evaluate advanced models such as:

- LightGBM
- XGBoost
- Deep Learning models such as Neural Networks

These methods may enhance predictive performance and capture more complex non-linear demand relationships.

4. Profit-Based Multi-Objective Optimization

The current optimization objective prioritizes **revenue maximization**. Future studies can extend the framework toward **multi-objective optimization**, balancing:

- Revenue maximization
- Profit maximization
- Market share retention
- Customer satisfaction

5. Real-Time Competitor Price Monitoring

Future systems can integrate **web scraping APIs and competitor intelligence systems** to continuously monitor competitor prices and automatically update price recommendations.

6. A/B Testing and Real-World Validation

Before enterprise deployment, future work should implement **controlled A/B testing** to validate whether optimized prices produce expected revenue gains under real market conditions.

16. References

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